Immutable Fuzz-Harness Submission

Tactix Development Project

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Outcome

The consolidation fuzz campaign is a terminal PASS. Claude Code reports 50,000/50,000 generated cases and 135/135 corpus replays with zero failures, 20,927 independently checked SAT models, 17,727 accepted native-LRAT proofs, complete differential agreement, and no sanitizer, crash, certificate, or scratch residue.

Two legs nevertheless ended with SLURM exit 127 after printing their complete, clean summaries. Claude localized this to a harness infrastructure defect, not solver behavior: a live edit shifted Bash's saved file offset while the script was blocked in its Python child. Fuzz submission now executes a read-only, content-hashed harness snapshot, preventing recurrence. No replacement fuzz campaign was necessary or submitted.

1 Cause

The old batch script invoked the live repository path `tools/fuzz/run.sh`. While jobs `1648215` and `1648217` waited for `fuzz.py`, commit `8a9c8f7` changed a comment in that file by 16 bytes. When Bash resumed after the child exited successfully, its saved byte offset now addressed shifted text and an option was executed as a shell command. This explains exit 127 after the already complete oracle summary.

The solver pins, cases, proofs, models, and oracle results remained exact. Claude therefore classified both tails as post-oracle infrastructure artifacts and the overall campaign as terminal PASS.

2 Repair

At submission, `tools/fuzz/slurm.sh` now creates a unique `build/pin-fuzz-harness-*` directory containing only the runner, Python harness modules, stress scenarios, and regression corpus used by the leg. It computes a deterministic aggregate SHA-256, makes the pin read-only, records the hash in the job log, and emits a batch body that executes only the pinned runner. A successful job removes the snapshot; a failed job retains it for Claude-owned diagnosis.

The pinned `run.sh` receives explicit repository and default-output roots. It therefore resolves solver and oracle paths exactly as before and preserves failure bundles outside the ephemeral read-only pin. A caller-supplied `-outdir` still takes precedence. Seeds, case counts, oracle semantics, and the 50-physical-core allocation policy are unchanged.

Development Plan §8.2 now requires immutable, hashed executed harness inputs for every fuzz leg and forbids batch execution of live repository scripts.

3 Verification

The new regression test submits through a mocked `sbatch` and verifies:

Property	Observed result
Three physical cores	six logical CPUs, 24,000 MiB, three workers
Executed runner	unique read-only harness pin
Live runner reference	absent from emitted batch
Snapshot identity	independent SHA-256 recomputation exact
Output roots	explicit repository and persistent failure directory

Shell syntax validation passes. The focused CTest set is 6/6: the new fuzz snapshot regression, Tier-D pinning, citations, section references, proof-rule synchronization, and the rule/policy boundary. The test removes its exact temporary pin and submits no cluster job.

4 Disposition

Accepted solver baseline `16d46d5` and its terminal fuzz PASS are closed. This repair is an isolated tooling increment; it changes no solver source. Convergence resumes directly at the next measured current-Tactix/current-Kissat broad search-quality gap.